

Arbitrary Poisson Boundary Measures Give Coordinate Multipliers

Candidate full-result packet for arXiv:2001.09616

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Status: full solution likely valid; human review required. For every finite positive Borel measure on the sphere, the associated Poisson-weighted volume measure is Carleson for the Dirichlet-type space in the source. Hence every coordinate function is a multiplier, and the multiplication tuple is a joint 2-isometry. The proof is analytic and does not assume polynomial density. Novelty confidence is moderate after a bounded search found no explicit later answer.

1 Source target and exact scope

The source is S. Chavan, R. Gupta, and Md. R. Reza, *Dirichlet-type spaces on the unit ball and joint 2-isometries*, arXiv:2001.09616v2 and J. Funct. Anal. 279 (2020), 108733 [1]. Let $d \geq 2$, let V be normalized volume measure on $\mathbb{B}^d$, and let σ be normalized surface measure on $\partial\mathbb{B}^d$. The source uses the Euclidean Poisson kernel for the real $2d$ -ball,

$$P(z, \eta) = \frac{1 - \|z\|^2}{\|z - \eta\|^{2d}}, \quad z \in \mathbb{B}^d, \quad \eta \in \partial\mathbb{B}^d, \quad (1)$$

and $P[\mu](z) = \int_{\partial\mathbb{B}^d} P(z, \eta) d\mu(\eta)$.

For a finite positive Borel measure μ on $\partial\mathbb{B}^d$, its scalar Dirichlet-type space consists of the holomorphic functions with

$$\|f\|_{\mathcal{D}(\mu)}^2 = \|f\|_{H^2(\mathbb{B}^d)}^2 + \frac{1}{d} \int_{\mathbb{B}^d} \|\nabla f(z)\|^2 P[\mu](z) dV(z) < \infty. \quad (2)$$

Published Proposition 3.5 says that all coordinate functions are multipliers if and only if

$$\int_{\mathbb{B}^d} |f(z)|^2 P[\mu](z) dV(z) \leq C_\mu \|f\|_{\mathcal{D}(\mu)}^2. \quad (3)$$

Page 12 proves this only when μ has bounded density and then states:

“We do not know whether the weighted volume measure with weight being the Poisson integral of an arbitrary finite positive Borel measure μ on $\partial\mathbb{B}^d$, $d \geq 2$ is a Carleson measure for $\mathcal{D}(\mu)$.”

The theorem below proves (3) for every such μ .

2 Proof intuition

The Poisson integral is linear in the boundary measure, so the decisive case is one atom δ_η . Its density looks highly singular in the usual coordinates. Center real polar coordinates at the singular boundary point η . The singular power cancels the polar Jacobian exactly:

$$P(z, \eta) dV(z) = c_{2d} (2\langle \omega, \eta \rangle_{\mathbb{R}} - \rho) d\rho d\omega, \quad z = \eta - \rho\omega.$$

The following provides a partial answer to Question [1.2\(a\)](#).

Corollary 3.7. *For a bounded measurable function $w : \partial\mathbb{B}^d \rightarrow [0, \infty)$, consider the weighted surface area measure μ_w with weight function w . Then the d -tuple $\mathcal{M}_z$ on $\mathcal{D}(\mu_w)$ is a joint 2-isometry.*

Proof. In view of Proposition [3.5](#) and Theorem [3.4](#), it suffices to check that there exists a positive constant C_w such that

$$\int_{\mathbb{B}^d} |f(z)|^2 P[\mu_w](z) dV(z) \leq C_w \|f\|_{\mathcal{D}(\mu_w)}^2, \quad f \in \mathcal{D}(\mu_w).$$

Since $\mathcal{D}(\mu_w) \subseteq H^2(\mathbb{B}^d) = \mathcal{H}_d$ and $\mathcal{H}_d$ is contractively embedded into $\mathcal{H}_{d+1}$ (see Example [3.1](#) and Remark [3.2\(i\)](#)) and $P[\mu_w]$ is bounded above by $\sup w$, we obtain the desired inequality with $C_w = \sup w$. $\square$

We do not know whether the weighted volume measure with weight being the Poisson integral of an arbitrary finite positive Borel measure μ on $\partial\mathbb{B}^d$, $d \geq 2$ is a Carleson measure for $\mathcal{D}(\mu)$.

4. ROTATION INVARIANT AND PLURIHARMONIC MEASURES

Figure 1: Source p. 12: the bounded-density corollary and the unresolved arbitrary-measure statement.

Thus the measure has effective radial dimension one at its tip. A weighted one-dimensional Hardy estimate controls the atomic weighted L^2 norm by the tip trace and the atomic gradient energy.

It remains to bound that trace. Average f over the unitary rotations that fix η . This preserves the radial value and contracts both the Hardy norm and the atomic energy, while reducing f to a one-variable function $h(\langle z, \eta \rangle)$. After the transverse variables are integrated out, the energy density of h' is at least $c_d/(1-r)$ on every dyadic nontangential box at 1. Submean estimates and a bounded-overlap dyadic sum then bound the trace. Integrating the resulting uniform atomic inequality against μ proves the full statement.

3 Atomic geometry and trace estimates

For $\eta \in \partial\mathbb{B}^d$ define, with extended values allowed,

$$\mathcal{E}_\eta(f) = \int_{\mathbb{B}^d} \|\nabla f(z)\|^2 P(z, \eta) dV(z), \tag{4}$$

$$\mathcal{I}_\eta(f) = \int_{\mathbb{B}^d} |f(z)|^2 P(z, \eta) dV(z). \tag{5}$$

Lemma 1 (Tip trace and anchored atomic Poincare inequality). *Let f be holomorphic on $\mathbb{B}^d$ and suppose $\mathcal{E}_\eta(f) < \infty$. Then f has a single finite-energy trace $\text{Tr}_\eta f$ at the tip η : along almost every*

inward ray from η the limit exists and equals the same number. Moreover,

$$\mathcal{I}_\eta(f) \leq 2|\mathrm{Tr}_\eta f|^2 + \frac{3}{2}\mathcal{E}_\eta(f). \quad (6)$$

Proof. Regard $\mathbb{C}^d$ as $\mathbb{R}^N$, $N = 2d$, and put $a = \langle \omega, \eta \rangle_{\mathbb{R}}$. The coordinates

$$z = \eta - \rho\omega, \quad a > 0, \quad 0 < \rho < 2a,$$

parametrize the ball up to null sets. Indeed,

$$1 - |z|^2 = \rho(2a - \rho), \quad |z - \eta| = \rho.$$

Since $dV = c_N \rho^{N-1} d\rho d\omega$, the exact cancellation is

$$P(z, \eta) dV(z) = c_N (2a - \rho) d\rho d\omega. \quad (7)$$

We first justify the common trace. On every compact connected angular set $K \Subset \{a > 0\}$ and for all sufficiently small ρ , the density in (7) is bounded below. The radial part of the energy implies that $f(\eta - \rho\omega)$ has a limit $L(\omega)$ for almost every $\omega \in K$, and convergence holds in $L^2(K)$. In centered polar coordinates the angular part of the Euclidean gradient is $\rho^{-1}\nabla_\omega$. Hence

$$\int_0^{\rho_0} \rho^{-2} \int_K |\nabla_\omega f(\eta - \rho\omega)|^2 d\omega d\rho < \infty.$$

There is a sequence $\rho_n \downarrow 0$ on which the inner angular energy tends to zero. The Poincaré inequality on K , followed by the $L^2(K)$ convergence, shows that L is constant on K . Exhausting the connected hemisphere $\{a > 0\}$ by overlapping compact sets gives one common value, denoted $\mathrm{Tr}_\eta f$.

Fix a direction and write $L = 2a$ and $u(\rho) = f(\eta - \rho\omega)$. The fundamental theorem of calculus and Cauchy–Schwarz give

$$|u(\rho) - \mathrm{Tr}_\eta f|^2 \leq \rho \int_0^\rho |u'(s)|^2 ds.$$

After multiplication by $L - \rho$ and integration, the coefficient of $|u'(s)|^2$ is

$$A_L(s) = \int_s^L \rho(L - \rho) d\rho.$$

Writing $t = s/L$, direct factorization gives

$$\frac{A_L(s)}{L^2(L - s)} = \frac{(1 - t)(2t + 1)}{6} \leq \frac{3}{16}.$$

Because $L \leq 2$,

$$\int_0^L (L - \rho) |u(\rho) - \mathrm{Tr}_\eta f|^2 d\rho \leq \frac{3}{4} \int_0^L (L - \rho) |u'(\rho)|^2 d\rho. \quad (8)$$

Integrate over ω , use (7), and note that $|u'| \leq \|\nabla f\|$. Since $\int_{\mathbb{B}^d} P(z, \eta) dV(z) = 1$, the inequality $|u|^2 \leq 2|\mathrm{Tr}_\eta f|^2 + 2|u - \mathrm{Tr}_\eta f|^2$ and (8) yield (6). $\square$

Lemma 2 (Uniform holomorphic tip trace). *For every $d \geq 2$ there is $C_d < \infty$ such that, for every $\eta \in \partial\mathbb{B}^d$ and every holomorphic f with $f \in H^2(\mathbb{B}^d)$ and $\mathcal{E}_\eta(f) < \infty$,*

$$|\mathrm{Tr}_\eta f|^2 \leq C_d (\|f\|_{H^2(\mathbb{B}^d)}^2 + \mathcal{E}_\eta(f)). \quad (9)$$

Proof. Unitary invariance reduces to $\eta = e_1$. Let $G \simeq U(d-1)$ be the compact group of unitaries fixing e_1 , and let

$$(Qf)(z) = \int_G f(Uz) dU.$$

Haar averaging and Jensen's inequality show

$$\|Qf\|_{H^2} \leq \|f\|_{H^2}, \quad \mathcal{E}_{e_1}(Qf) \leq \mathcal{E}_{e_1}(f). \quad (10)$$

The holomorphic G -invariants are precisely the functions of the first coordinate, so $Qf(z) = h(z_1)$, where $h(\lambda) = f(\lambda, 0, \dots, 0)$. The tip trace is invariant under G and linear on the finite-energy class; hence $\text{Tr}_{e_1} Qf = \text{Tr}_{e_1} f$.

Put $A = 1 - |\lambda|^2$ and $b = |1 - \lambda|^2$. Integrating the other $d-1$ complex coordinates in polar coordinates gives, up to a positive dimensional normalization,

$$\mathcal{E}_{e_1}(Qf) = c_d \int_{\mathbb{D}} |h'(\lambda)|^2 W_d(\lambda) dA(\lambda), \quad W_d(\lambda) = \int_0^A \frac{(A-u)u^{d-2}}{(b+u)^d} du. \quad (11)$$

We record the only property of W_d needed below. Let $\delta_n = 2^{-n}$ and let J_n be the real interval $[1 - \delta_n, 1 - \delta_n/2]$. Choose fixed, sufficiently small enlargements $Q_n \Subset \mathbb{D}$ of the δ_n -neighborhoods of J_n . On Q_n ,

$$A \asymp \delta_n, \quad b \asymp \delta_n^2. \quad (12)$$

Restricting the integral in (11) to $b \leq u \leq 2b$ gives

$$W_d(\lambda) \geq \frac{c'_d}{\delta_n}, \quad \lambda \in Q_n. \quad (13)$$

Indeed, on that interval the integrand is bounded below by a dimensional constant times $\delta_n b^{-2}$, and the interval has length b .

For each $r \in J_n$, the submean inequality for the analytic function h' on a disk of radius comparable to δ_n gives

$$|h'(r)|^2 \leq C \delta_n^{-2} \int_{Q_n} |h'(\lambda)|^2 dA(\lambda).$$

Combining this with (13) and then integrating over J_n yields

$$\int_{J_n} |h'(r)|^2 dr \leq C_d \int_{Q_n} |h'(\lambda)|^2 W_d(\lambda) dA(\lambda) =: C_d E_n. \quad (14)$$

The boxes Q_n have bounded overlap. Therefore

$$\begin{aligned} \int_{1/2}^1 |h'(r)|^2 dr &\leq \sum_n |J_n|^{1/2} \left(\int_{J_n} |h'(r)|^2 dr \right)^{1/2} \\ &\leq C_d \left(\sum_n \delta_n \right)^{1/2} \left(\sum_n E_n \right)^{1/2} \leq C_d \mathcal{E}_{e_1}(Qf)^{1/2}. \end{aligned}$$

Thus $h(r)$ has a finite limit at 1. The same submean estimate on slightly smaller boxes controls the oscillation in every fixed Stolz cone, so this limit is the tip trace. Finally, bounded point evaluation at $e_1/2$ in $H^2(\mathbb{B}^d)$ gives

$$|\text{Tr}_{e_1} f| \leq |h(1/2)| + C_d \mathcal{E}_{e_1}(Qf)^{1/2} \leq C_d (\|Qf\|_{H^2} + \mathcal{E}_{e_1}(Qf)^{1/2}).$$

Use (10) and square the estimate. □

4 Full affirmative solution

Theorem 1 (Arbitrary Poisson measures are Carleson). *Let $d \geq 2$ and let μ be any finite positive Borel measure on $\partial\mathbb{B}^d$. Then there is a dimensional constant C_d such that every $f \in \mathcal{D}(\mu)$ satisfies*

$$\int_{\mathbb{B}^d} |f(z)|^2 P[\mu](z) dV(z) \leq C_d \left(\mu(\partial\mathbb{B}^d) \|f\|_{H^2(\mathbb{B}^d)}^2 + \int_{\mathbb{B}^d} \|\nabla f(z)\|^2 P[\mu](z) dV(z) \right). \quad (15)$$

Consequently $P[\mu]dV$ is a Carleson measure for $\mathcal{D}(\mu)$, every coordinate function z_j is a multiplier of $\mathcal{D}(\mu)$, and the tuple of coordinate multipliers is a joint 2-isometry.

Proof. Tonelli's theorem applied to (1) gives

$$\int_{\mathbb{B}^d} |f|^2 P[\mu] dV = \int_{\partial\mathbb{B}^d} \mathcal{I}_\eta(f) d\mu(\eta), \quad (16)$$

$$\int_{\mathbb{B}^d} \|\nabla f\|^2 P[\mu] dV = \int_{\partial\mathbb{B}^d} \mathcal{E}_\eta(f) d\mu(\eta). \quad (17)$$

The second quantity is finite for $f \in \mathcal{D}(\mu)$, so $\mathcal{E}_\eta(f) < \infty$ for μ -almost every η . Lemmas 1 and 2, with constants independent of η , give

$$\mathcal{I}_\eta(f) \leq C_d (\|f\|_{H^2(\mathbb{B}^d)}^2 + \mathcal{E}_\eta(f))$$

for almost every η . Integrating this inequality and using (16)–(17) proves (15). In particular,

$$\int_{\mathbb{B}^d} |f|^2 P[\mu] dV \leq C_d \max\{\mu(\partial\mathbb{B}^d), d\} \|f\|_{\mathcal{D}(\mu)}^2.$$

This is exactly published Proposition 3.5(ii), so its equivalence proves that all z_j are multipliers. Published Theorem 3.4 then says that their tuple is a joint 2-isometry. $\square$

Remark 1. The proof works directly for arbitrary $f \in \mathcal{D}(\mu)$. It does not pass through polynomials, radial dilations, or an unproved density statement. The argument is scalar, matching Question 1.2(a). An operator-valued semispectral extension would require a separate noncommutative trace lemma and is not claimed here.

5 Adversarial verification

Step	Verdict	Check
Atomic disintegration	Valid	The power ρ^{1-2d} in P cancels the real $2d$ -dimensional polar Jacobian exactly, giving (7).
Tip trace	Valid	Radial energy gives ray limits; the ρ^{-2} angular energy and the Poincare inequality on compact angular sets force a single common value.
Anchored Hardy bound	Valid	The scalar kernel factors as $A_L(s)/(L^2(L-s)) = (1-t)(2t+1)/6 \leq 3/16$.
Stabilizer averaging	Valid	Haar averaging fixes the radial restriction, preserves the atomic weight, and contracts both quadratic norms by Jensen.
Marginal estimate	Valid	On dyadic Stolz boxes, restricting (11) to $u \in [b, 2b]$ gives $W_d \gtrsim \delta^{-1}$ uniformly in scale.
Arbitrary functions	Valid	Tonelli is applied to the actual finite Dirichlet energy; no polynomial approximation or boundary value with respect to μ is assumed.
Passage to multipliers	Valid	The final estimate is precisely the condition already proved equivalent to coordinate multiplication in source Proposition 3.5.

The included script `code/verify_atomic_geometry.py` is a guard against algebraic and scaling mistakes, not part of the proof. It checks 800 centered-polar identities, 10,001 samples of the exact Hardy kernel, and 567 dyadic marginal samples in complex dimensions 2 through 8. From the packet directory, run:

```
conda run -no-capture-output -n sandbox python code/verify_atomic_geometry.py
```

6 Novelty check, limitations, and recommendation

Bounded literature search. The four lightweight indexes for run `fa_banach_001` contained no entry for arXiv:2001.09616 or this question. External searches used the exact unresolved sentence, the exact title with `multiplier`, `Poisson integral`, and `Carleson measure`, the published problem/proposition labels, and the later visible Dirichlet-type papers and publication lists of the three source authors. The search found the original JFA paper, papers on standard radial Dirichlet scales, and later work on the bidisc, but no explicit answer to the arbitrary Euclidean-Poisson-measure question. This was a bounded search, not an exhaustive priority claim.

Limitations. The result answers the scalar arbitrary-measure problem exactly as posed. It does not claim an operator-valued semispectral analogue, an invariant-Poisson analogue, sharp constants, compactness of the embedding, or multiplier characterizations beyond the coordinate functions.

Recommendation. The mathematical verdict is **likely valid**, with high proof confidence. Send to human review. The most important checks are the finite-energy tip trace argument in Lemma 1 and the identification of the stabilizer-averaged nontangential trace in Lemma 2. The main bibliographic task is a citation search for later papers that cite the 2020 JFA article and address Question 1.2(a).

References

- [1] S. Chavan, R. Gupta, and Md. R. Reza, *Dirichlet-type spaces on the unit ball and joint 2-isometries*, J. Funct. Anal. 279 (2020), no. 10, 108733; arXiv:2001.09616v2.